V01.03 – Mirror Theory III

Recursive Observerhood in Context

A Comparative Computational Framework for Cognition, Consciousness and Artificial Intelligence

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Abstract

Mirror Theory I proposed a minimal computational account of recursive observerhood. It treated observerhood not as a primitive subject, soul, point of view or phenomenological datum, but as a derived regime of bounded viability-constrained systems that maintain compressed models of the world, compressed models of their own internal state, and models of the reliability of those self-models. Mirror Theory II developed the philosophical consequences of that result for identity, persistence, death, meaning, objectivity and realism. The present paper situates Mirror Theory within the broader landscape of contemporary cognitive science, philosophy of mind and artificial intelligence.

The central claim is comparative rather than foundational. Mirror Theory is not presented as a replacement for predictive processing, active inference, global workspace theory, integrated information theory, autopoiesis, enactivism, self-model theory, intentional systems theory or strange-loop accounts. It is presented as a candidate bridge among them: a structural account of the conditions under which a system becomes an observer at all. Its distinctive contribution is to make observerhood the explanatory target rather than an assumed starting point. It asks not primarily what consciousness feels like, how information becomes globally available, how organisms regulate action, or whether a system has integrated cause-effect power, but what minimal organisation allows a bounded system to maintain a world-model in which it is itself operationally located and recursively correctable.

The paper argues that Mirror Theory occupies a middle position. Against naive realism, it denies that observers passively receive a ready-made world without constructive modelling. Against radical idealism and simulation-first metaphysics, it insists that observerhood remains constrained by structures not freely invented by the observer. Against purely behavioural or functional descriptions, it requires explicit recursive self-modelling and reliability-tracking. Against consciousness-first frameworks, it treats phenomenal consciousness as a further question rather than an initial primitive. The paper concludes by proposing a research programme: develop operational metrics for recursive self-modelling, specify minimal architectures for observerhood, compare Mirror Theory against active inference and self-model theory in artificial agents, and identify failure cases in which prediction, self-reference or feedback are present but observerhood is not.

Keywords: Mirror Programme; Mirror Theory; observerhood; recursive observerhood; cognition; consciousness; artificial intelligence; predictive processing; active inference; self-model theory; computational cognition.

Contents

Roadmap Position

This paper is **V01.03 – Mirror Theory III**, the comparative closing paper of the first Theory Arc in Volume I of the Mirror Programme. V01.01 first derives recursive observerhood as a computational regime. V01.02 then develops its philosophical consequences. V01.03 locates the framework in relation to neighbouring traditions in cognitive science, philosophy of mind and artificial intelligence. Its role is not to replace those traditions, but to clarify the explanatory niche of Mirror Theory: observerhood as an intermediate target between adaptive regulation and full phenomenal consciousness.

1 Introduction

Many theories in cognitive science and philosophy of mind begin after an observer has already entered the room. A subject perceives. An organism acts. A brain predicts. A workspace broadcasts. An integrated system is conscious. A self-model becomes transparent. An intentional system is treated as having beliefs and desires. Each approach may illuminate something essential, but the explanatory role of the observer is often already present in the formulation.

Mirror Theory begins one step earlier. It asks:

What kind of organisation must exist before the term observer can be applied to a system in a non-metaphorical, computationally meaningful way?

The answer developed in Mirror Theory I was deliberately narrow. Observerhood was defined as a derived computational regime in bounded viability-constrained systems. The derivation began with four primitive assumptions: distinguishability, dynamics, viability and bounded computational capacity. From these it attempted to derive persistence, token formation, compression, prediction, predictive representation, self-modelling and recursive self-modelling. The central theorem stated that where future viability depends on compressed internal state, and where the reliability of that internal model carries additional information about future viability, recursive self-modelling reduces optimal predictive risk under log loss and is favoured when its viability value exceeds its computational cost [21].

Mirror Theory II then asked what follows if observerhood is derived rather than assumed. Its central claim was interpretive: if observerhood is a maintained recursive regime, then identity is persistence of observer-maintaining constraints; meaning is instantiated when symbolic or representational structures matter for the system's continued organisation; objectivity is convergent constraint-tracking across situated observers; and realism must explain not only what exists, but how observerhood can arise within what exists [22].

The present paper has a different task. It does not add another theorem. It does not claim to solve phenomenality. It does not attempt to replace the large existing literature on consciousness, cognition, agency or artificial intelligence. Its task is to locate Mirror Theory among those literatures and to make explicit what it contributes, what it borrows, what it rejects, and what remains open.

This positioning task matters because Mirror Theory sits close to several established traditions. Predictive processing treats perception as inference under hierarchical generative models, often described as the minimisation of prediction error [19, 6]. The free-energy principle and active inference treat perception, action and learning under a variational optimisation framework for self-organising systems [11, 12]. Autopoiesis treats living systems as self-producing autonomous unities [15]. Enactivism and sensorimotor theory stress that cognition and perception are enacted through embodied engagement with the world [24, 17]. Global workspace theory explains conscious access through global availability or broadcasting within an architecture [2, 3, 8]. Integrated Information Theory begins from phenomenology and seeks physical postulates corresponding to the intrinsic structure of experience [23, 18, 1]. Self-model theory treats the experienced self as a transparent phenomenal model rather than a metaphysical subject [16]. Dennett's intentional stance treats beliefs and desires as predictive attributions made from a particular explanatory strategy [9]. Hofstadter's strange-loop account emphasises recursive self-reference as central to selfhood [13].

Given that landscape, the most important question is not whether Mirror Theory sounds new. The important question is whether it does explanatory work not already performed by these approaches. This paper's answer is conditional and modest: Mirror Theory may be useful if observerhood is treated as an intermediate explanatory target. It is not identical with consciousness, agency, life, selfhood, information integration, prediction or embodiment. It is the proposed structural regime in which a bounded system becomes capable of maintaining a world-model that includes the system itself as a reliability-sensitive component.

In compressed form:

> *Predictive processing explains prediction. Active inference explains perception-action regulation. Autopoiesis explains self-producing living organisation. Enactivism explains embodied sense-making. Global workspace theory explains access and availability. Integrated Information Theory explains consciousness in terms of intrinsic cause-effect structure. Self-model theory explains the phenomenal self-model. Mirror Theory asks what computational organisation must exist for observerhood to be derived rather than assumed.*

This paper therefore functions as a bridge. It situates Mirror Theory within existing work while preserving its narrower target: recursive observerhood.

2 Scope and negative commitments

The most important way to misread Mirror Theory is to inflate it. The theory is not a universal theory of everything, a replacement for physics, a completed neuroscience, a proof of machine consciousness, a panpsychist doctrine, a simulation argument, or a metaphysical declaration that reality is mental. It is a proposed formal and philosophical account of observerhood as a derived organisational regime.

The following negative commitments are therefore essential.

2.1 Mirror Theory is not a theory of phenomenal consciousness in full

Mirror Theory distinguishes observerhood from phenomenality. A system may satisfy many structural conditions for recursive observerhood without thereby establishing that there is something it is like to be that system. This is not a minor footnote. It is a boundary condition. The hard problem of consciousness, as classically formulated by Chalmers [5], is not dissolved by renaming it. Mirror Theory attempts to explain a structural prerequisite for world-bearing systems: persistence, modelling, self-inclusion and recursive correction. It does not claim that this structural account exhausts subjective experience.

This distinction also separates Mirror Theory from consciousness-first accounts. Integrated Information Theory begins with the existence and structure of experience and attempts to infer physical postulates corresponding to that structure [23, 18, 1]. Mirror Theory begins before that. It asks what organisation allows a system to become an observer to whom anything can appear as a world. This may be relevant to consciousness, but it is not identical to it.

2.2 Mirror Theory is not panpsychism

Mirror Theory does not claim that all systems have experience or observerhood. A thermostat distinguishes a temperature threshold. A bacterium may move along a chemical gradient. A rock persists. A whirlpool maintains structure. None of these facts is sufficient for Mirror observerhood. The relevant architecture requires maintained world-modelling, self-inclusion within that model, and some capacity to track and correct the reliability of that self-inclusion over time.

This is why Mirror Theory must resist both inflation and collapse. If every difference-responsive system is an observer, the concept becomes empty. If only linguistically reflective human beings count as observers, the theory smuggles in too much. The target is a middle region: systems whose organisation is rich enough to include self-modelling and recursive reliability-tracking, but whose status does not depend on assuming human phenomenology at the outset.

2.3 Mirror Theory is not simulation theory

Simulation arguments ask whether the apparent universe might be generated by some deeper computational substrate [4]. Mirror Theory asks a different question. It does not say that the universe is artificial or externally rendered. It says that an observer's access to any universe, whether base-level or simulated, is mediated by modelling, compression and correction. Simulation theory makes the ontology of the world artificial. Mirror Theory makes observation constructive.

The distinction is simple: simulation theory asks whether the territory is generated; Mirror Theory asks how any territory becomes available as a world to an observer. A real mountain is not made fake by the fact that a hiker carries only a map. The map is constructed, but the mountain constrains it. Likewise, Mirror Theory treats experienced worldhood as constructed under constraint, not as unconstrained invention.

2.4 Mirror Theory is not eliminativism

Mirror Theory does not eliminate observers by reducing them to meaningless physical happenings. It relocates them. An observer is not a primitive witness inserted into nature from outside, but neither is it an illusion to be dismissed. It is a real organisational regime, if the conditions for that regime are met. This is close in spirit to some naturalistic approaches in philosophy of mind, including Dennett's rejection of Cartesian spectators and his emphasis on patterns, functions and explanatory stances [9, 10]. But Mirror Theory adds a constructive requirement: the observer predicate should not merely be attributed when useful; it should be derived from a sequence of formal dependencies.

2.5 Mirror Theory is not a finished empirical theory

The theory remains incomplete unless it generates operational tests. Mirror Theory I proposed information-theoretic and computational tests, but those are not yet an empirical science. The present paper therefore treats Mirror Theory as a research programme. Its value lies not in having already answered every question, but in specifying where the questions should be asked.

3 Minimal reconstruction of Mirror Theory

This section summarises the core architecture that Mirror Theory I and Mirror Theory II developed. The summary is intentionally compressed. The purpose is not to reproduce the formal derivation but to make clear what is being compared with the surrounding literature.

3.1 Primitive assumptions

Mirror Theory I begins with four primitive assumptions.

1. **Distinguishability.** A system must be able, in some formal sense, to distinguish states or differences. Without distinguishability, no token, model or representation can be derived.
2. **Dynamics.** States must transition over time or under an ordering relation. Without dynamics, there is no persistence, prediction or correction.
3. **Viability.** Some states or trajectories must matter for the continued organisation of the system. Viability is not moral value; it is organisational preservation.
4. **Bounded computational capacity.** The system cannot encode or process the entire state space. It must compress, select and discard.

The methodological rule is that symbols, meanings, selves and observers may not be introduced as primitive entities. They must be derived or identified as failure points.

3.2 Derived sequence

The proposed dependency sequence is:

$$\begin{aligned}(\delta, \Delta, V, \kappa) &\to \text{persistence} \to \text{tokens} \to \text{compression} \\ &\to \text{prediction} \to \text{representation} \to \text{self-model} \\ &\to \text{recursive observerhood.}\end{aligned} \tag{1}$$

The sequence is not claimed to apply to every system. It is conditional: where the information, viability and capacity conditions are satisfied, recursive observerhood becomes a selected regime.

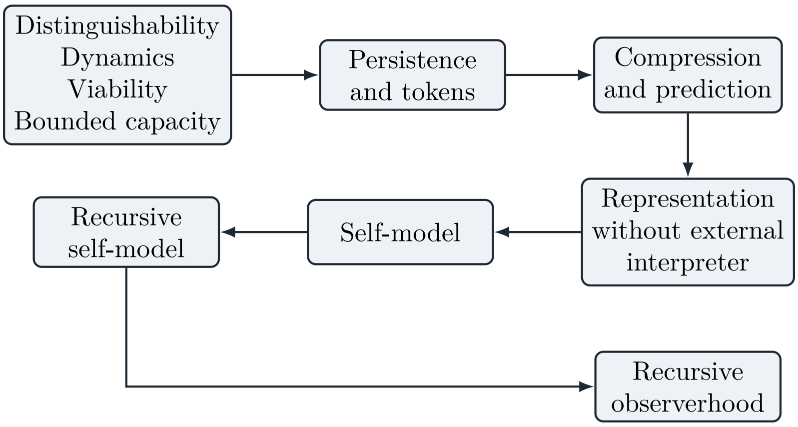

Figure 1: The compressed dependency chain proposed by Mirror Theory. The arrows should be read as derivational dependencies under stated conditions, not as a universal developmental sequence.

3.3 The core definition

Definition 3.1 (Recursive observerhood)**.** A recursive observer is a bounded viability-constrained system that maintains compressed models of its environment, compressed models of its own internal state, and models of the reliability of those self-models, such that recursive self-modelling contributes to the preservation, prediction or correction of the system's continued organisation.

This definition is intentionally structural. It does not say that all such systems are phenomenally conscious. It says that a certain observerhood predicate can be applied when a system maintains a world-model in which its own modelling activity becomes an operationally relevant and correctable component.

3.4 The formal contribution of Mirror Theory I

Mirror Theory I's central formal result can be stated informally as follows. If future viability depends on compressed internal state, and if the reliability or error-profile of the self-model carries additional information about future viability, then a recursive architecture that models self-model reliability achieves lower optimal predictive risk under log loss than an otherwise comparable architecture lacking that variable. A threshold theorem then states that, under a viability selection rule, recursive observerhood becomes selected once the viability value of modelling self-model reliability exceeds its computational cost [21].

This is important because Mirror Theory does not say that recursion is magical. Recursion is expensive. It is selected only when the world, the body or the system's own internal dynamics make self-model reliability relevant to future viability. If a system's internal state is not viability-relevant, self-modelling is not derived. If the reliability of the self-model is not viability-relevant, recursive self-modelling is not derived. If computational cost exceeds predictive benefit, recursive observerhood is not selected.

3.5 The philosophical contribution of Mirror Theory II

Mirror Theory II interpreted observerhood as a maintained recursive organisation rather than a primitive subject. It proposed that identity is path-dependent persistence of observer-maintaining constraints; death is collapse of the recursive persistence constitutive of observer identity; objectivity is convergence among situated observers under shared constraints; and meaning is instantiated when representational structures become functionally significant for the system's continued organisation [22].

This is the philosophical background against which the present paper asks: where does Mirror Theory sit among existing frameworks?

4 A map of the surrounding landscape

Theories of cognition and consciousness differ because they ask different primary questions. The differences are often obscured when every theory is forced into the single question, "What is consciousness?" Mirror Theory is better evaluated by asking what explanatory target each theory places at the centre.

Table 1: Primary explanatory targets in the surrounding theoretical landscape.

Framework	**Primary question**	**Typical explanatory object**	**Mirror Theory relation**
Predictive processing	How does perception and cognition arise through prediction and error correction?	Hierarchical generative models; prediction error minimisation	Adopts modelling and prediction but asks when prediction becomes recursively self-locating observerhood.
Active inference / free-energy principle	How do self-organising systems regulate perception and action under variational principles?	Agent-environment dynamics; expected free energy; action-perception loops	Compatible with active inference but isolates recursive observerhood as a specific regime within adaptive regulation.
Autopoiesis	What makes a living system an autonomous self-producing unity?	Organisational closure; self-production; autonomy	Inherits the importance of self-maintenance but does not equate life with observerhood.
Enactivism / sensorimotor theory	How is cognition enacted through embodied activity?	Sensorimotor contingencies; embodied action; world-making through engagement	Agrees that worldhood is enacted but adds a formal requirement for recursive self-model reliability.
Global workspace theory	How does information become consciously accessible and reportable?	Global broadcasting, access, integration across specialist processors	Does not explain access consciousness directly; asks what prior observer architecture could make global availability meaningful for a system.
Integrated Information Theory	What physical systems are conscious, to what degree, and in what way?	Intrinsic cause-effect structures; integrated information	Differs by beginning from observerhood rather than phenomenology.
Self-model theory	How does the self appear as a transparent phenomenal model?	Phenomenal self-models; transparency; ownership	Treats self-modelling as necessary but adds recursive reliability-tracking and viability selection.
Intentional stance	When is it useful to predict a system by attributing beliefs and desires?	Predictive interpretation of agents	Seeks constructive conditions for observerhood rather than only an interpretive stance.

Framework	**Primary question**	**Typical explanatory object**	**Mirror Theory relation**
Strange-loop accounts	How does selfhood emerge from recursive self-reference?	Self-reference, loops, symbol systems	Agrees on recursion but distinguishes mere self-reference from viability-constrained recursive self-modelling.

The table suggests a simple conclusion: Mirror Theory should not compete with every theory at the same level. Its niche is not "another consciousness theory" but a candidate theory of observerhood. It asks when a system becomes the kind of thing for which perception, action, access, integration, selfhood and meaning can be organised around a maintained point of view.

5 Predictive processing and the Bayesian brain

Predictive processing is among the closest neighbours of Mirror Theory. Classical predictive coding models treat perception as a process in which higher levels send predictions downward while lower levels send residual prediction errors upward [19]. In later philosophical and cognitive-scientific formulations, the brain is described as a prediction machine that uses hierarchical generative models to infer the hidden causes of sensory input [6, 14]. The free-energy principle generalises this tendency by placing perception, learning and action within a broader optimisation framework [11, 12].

Mirror Theory agrees with three core ideas from predictive processing. First, a system does not passively copy the world. It models hidden structure from limited signals. Second, perception is inseparable from compression: no bounded system can encode the whole world in full detail. Third, prediction matters because models are tested by what they allow the system to anticipate, correct and do.

However, Mirror Theory differs in explanatory target. Predictive processing can describe systems that predict without becoming observers in the Mirror sense. A thermostat may predictably regulate temperature under simple dynamics. A machine-learning model may forecast time series. A visual cortex may reduce prediction error. None of this alone yields recursive observerhood. Mirror Theory asks when prediction becomes self-involving: when the system models not only external states but also its own modelling state, and then models the reliability of that self-model.

Claim 5.1 (Prediction is not yet observerhood)**.** Prediction becomes observer-relevant when the reliability of the system's own modelling activity becomes an operational variable for future viability. Without that step, prediction may be powerful but not yet recursive observerhood.

This claim does not reject predictive processing. It specifies a subcase. Mirror Theory can be read as adding a recursive observerhood condition to predictive architectures. If predictive processing explains how systems infer hidden causes of input, Mirror Theory asks when the inferring system must include itself, and the reliability of its own inference, within the model.

6 Active inference and the free-energy principle

Active inference extends predictive ideas into action. Under active inference, organisms do not merely update beliefs to fit sensory input; they also act to make sensory input conform to predictions, preferences or expected free-energy minimisation [11, 12]. The framework is attractive because it treats perception and action as coupled parts of a single regulatory process.

Mirror Theory is highly compatible with this picture. Its viability condition resembles, at an abstract level, the idea that self-organising systems must preserve themselves within certain bounds. Its compression and prediction requirements also resonate with generative modelling. Its rejection of passive spectator realism aligns with the active-inference claim that agents are not detached observers but systems embedded in action-perception loops.

The difference is again one of emphasis. Active inference is broader than Mirror Theory. It can describe homeostatic regulation, perception, motor control, exploration and policy selection. Mirror Theory focuses on the narrower question of when such a system becomes an observer through recursive self-modelling. A simple active-inference agent may maintain beliefs about hidden environmental states and act to regulate its sensory flow. But Mirror observerhood requires that the agent also maintain models of its own internal modelling state, including reliability, error or distortion in that self-model.

Proposal 6.1 (Mirror-active inference interface)**.** Mirror observerhood can be investigated as a class of active-inference architectures in which the generative model explicitly includes (i) hidden environmental states, (ii) hidden internal bodily or operational states, and (iii) reliability variables governing the accuracy or distortion of the system's own self-model.

This proposal gives Mirror Theory empirical traction. It suggests that one should not merely ask whether an artificial or biological agent minimises prediction error or expected free energy. One should ask whether the agent's own self-modelling reliability becomes a variable in policy selection, learning or viability preservation.

7 Autopoiesis and organisational closure

Autopoiesis, developed by Maturana and Varela, treats living systems as self-producing autonomous unities [15]. A living system maintains itself by producing the components and boundaries that constitute it. Cognition, on this view, is deeply tied to life and autonomy rather than detached representation.

Mirror Theory shares the emphasis on organisation rather than substance. Both reject the idea that the relevant unit is simply a heap of matter. Both stress maintained organisation across change. Both treat boundaries and self-maintenance as central.

The difference is that Mirror Theory does not equate life, autonomy or self-production with observerhood. Autopoiesis may be necessary for many biological observers, but it is not sufficient for recursive observerhood. A cell may be self-producing without maintaining a recursive self-model. Conversely, Mirror Theory leaves open the possibility that non-biological systems could instantiate observerhood if they satisfy the relevant organisational conditions. The biological origin of many observers is therefore not denied, but substrate restriction is not built into the definition.

Autopoiesis explains how a system maintains itself as a living unity. Mirror Theory asks when a self-maintaining system, biological or artificial, also maintains a model of itself as a modelling system under reliability-sensitive correction.

This distinction matters for artificial intelligence. A future artificial system might lack biological autopoiesis but still maintain recursive self-models. Whether that is sufficient for observerhood depends on whether its self-models are operationally coupled to persistence, prediction, correction and viability. Mirror Theory therefore cannot simply import life as the criterion.

8 Enactivism and sensorimotor accounts

Enactivism argues that cognition is not inner representation alone but embodied action in a world. Varela, Thompson and Rosch's enactive approach stresses that cognition is a history of structural coupling between organism and environment [24]. Sensorimotor accounts of vision similarly emphasise mastery of sensorimotor contingencies rather than the construction of inner pictures alone [17].

Mirror Theory strongly agrees that worldhood is not passive reception. A world appears to a system through the differences it can register, the actions it can take, the consequences it can encounter and the corrections it can undergo. The observer is not outside the world; it is a pattern within the world maintaining a relation to world, model and self.

Where Mirror Theory differs is in its explicit formal requirement for recursive self-model reliability. Enactivism can explain sense-making through embodied engagement, but not all embodied engagement produces recursive observerhood. A plant bends toward light. A bacterium follows a gradient. An animal navigates affordances. These may be enactive in broad senses, but Mirror Theory reserves observerhood for systems whose model includes operational self-location and correction of self-model reliability.

In other words, enactivism explains how worldhood is enacted; Mirror Theory adds a condition for when the enacting system becomes an observer of its own world-model.

9 Global workspace theory

Global workspace theory explains conscious access through a cognitive architecture in which information becomes globally available to multiple specialist systems [2, 3]. Global neuronal workspace theory develops this idea neuroscientifically, emphasising access, reportability and large-scale availability [8, 7].

Mirror Theory does not deny the importance of global availability. For complex human cognition, recursively self-modelled information may often require wide availability across memory, attention, action, language and planning systems. A self-model that cannot influence action, correction or future prediction would be inert. In that sense, global broadcasting may be an implementation pathway for sophisticated observerhood.

However, the two theories ask different questions. Global workspace theory asks when information becomes conscious or reportable. Mirror Theory asks what organisation allows a system to be an observer at all. A workspace could broadcast information without itself explaining how the system came to maintain a world-model that includes itself as a reliability-sensitive component. Conversely, a primitive recursive observer might not have a human-like global workspace or linguistic reportability.

The relationship is therefore complementary. Global workspace theory may describe one architecture for high-level conscious access. Mirror Theory supplies a possible prior condition: the system must already have a self-involving model whose contents can matter for correction, action and persistence.

10 Integrated Information Theory

Integrated Information Theory is one of the most prominent contemporary theories of consciousness. In its classical formulation, IIT begins from axioms concerning the structure of experience and develops corresponding physical postulates. It attempts to determine which physical systems are conscious, to what degree and in what way, often through measures of integrated information or irreducible cause-effect power [23, 18, 1].

Mirror Theory differs sharply in starting point. IIT begins from consciousness or experience. Mirror Theory begins from observerhood as a derived organisation. IIT asks what physical substrate can account for the intrinsic structure of experience. Mirror Theory asks what computational structure allows a bounded system to become an observer before phenomenality is assumed.

There is nevertheless an important convergence. Both theories reject simple behavioural criteria. Both insist that internal organisation matters. Both resist reducing consciousness or observerhood to external input-output behaviour. Both are formal in aspiration.

The key difference is phenomenological priority. IIT is consciousness-first; Mirror Theory is observerhood-first. That does not make one approach right and the other wrong. It makes them address different layers. If IIT succeeds, it may explain what makes a physical system conscious. If Mirror Theory succeeds, it may explain what makes a system an observer capable of maintaining a world-model. The two could eventually constrain each other. A system might satisfy Mirror observerhood without satisfying IIT consciousness, or vice versa. Those divergence cases would be theoretically illuminating.

11 Self-model theory

Thomas Metzinger's self-model theory argues that the self is not a thing but a model, and that the phenomenal self appears as a transparent representational structure [16]. This is one of Mirror Theory's closest philosophical relatives. Both reject a primitive inner subject. Both treat selfhood as a constructed or modelled phenomenon. Both seek to explain why the self appears as something more substantial than it is.

Mirror Theory differs by making self-modelling part of a broader derivational chain. Self-modelling is necessary but not sufficient. A system must also maintain a world-model, preserve organisation under viability constraints, and track the reliability of its self-model over time. Mirror Theory is therefore less concerned with the phenomenology of transparency and more concerned with the computational conditions under which self-modelling becomes selected.

Self-model theory explains how the self appears as a model. Mirror Theory asks why a system would need such a model, when that need becomes recursive, and how that recursion contributes to observerhood.

The relationship is promising. Metzinger provides a rich account of the phenomenal self-model. Mirror Theory can provide an abstract account of why self-modelling and self-model reliability might be selected in viability-constrained systems.

12 Dennett, intentional systems and strange loops

Dennett's intentional stance treats beliefs and desires as part of a predictive strategy. We interpret a system as rational, attribute beliefs and desires, and use those attributions to predict behaviour [9]. Dennett's broader naturalism also resists Cartesian inner theatres and primitive spectators [10].

Mirror Theory is sympathetic to this anti-Cartesian impulse. It agrees that the observer should not be imagined as a little witness inside the head. It also agrees that high-level patterns can be real when they support explanation and prediction.

The difference is that Mirror Theory is not merely stance-relative. It does not say only that observerhood is a useful attribution made by external interpreters. It asks whether internal organisational conditions can make the observer predicate apply constructively. A chess program may be treated intentionally in some contexts; a thermostat can be described as "wanting" to keep the room warm; but Mirror Theory does not grant observerhood to any system for which intentional vocabulary is predictively useful. The system must maintain its own self-involving model and reliability-sensitive correction.

Hofstadter's strange-loop account is another close neighbour. It treats selfhood as emerging from self-referential symbolic loops [13]. Mirror Theory agrees that recursion matters, but it distinguishes recursion from observerhood. A self-referential sentence is recursive but not an observer. A feedback loop is recursive in a weak sense but not an observer. A quine refers to its own code but has no viability-constrained self-model. Mirror recursion must be operational: the self-reference must alter prediction, correction, persistence or action for the system itself.

13 Simulation arguments and constructed worldhood

Simulation theory asks whether our universe may be generated by a deeper computational substrate [4]. Mirror Theory does not require that claim. It is compatible with either base reality or simulated reality, because its main question is substrate-neutral: how does any structured reality become a world for an observer?

The distinction can be stated in one line:

> *Simulation theory makes the universe artificial; Mirror Theory makes observation constructive.*

This does not mean that Mirror Theory denies objective constraint. On the contrary, it treats reality as what corrects the model. An observer lives through a constructed world-model, but the model is constrained by structures beyond the observer's private interpretation. In ordinary terms: the wall is real because it does not obey the model when the model says the wall is absent.

This position avoids two mistakes. The first is naive realism, which imagines that observers simply receive the world exactly as it is. The second is unconstrained constructivism, which imagines that because experience is constructed, reality is arbitrary. Mirror Theory holds the middle position: worldhood is constructed under constraint.

14 What Mirror Theory contributes

The comparative analysis suggests that Mirror Theory contributes five things.

14.1 Observerhood as an explicit target

Most theories target consciousness, cognition, agency, perception, access, autonomy or selfhood. Mirror Theory targets observerhood. This matters because observerhood is often assumed in the background of those other concepts. If observerhood can be derived, then theories of consciousness and cognition gain a prior structural layer.

14.2 A derivational rule against smuggling in the observer

The methodological rule is severe: no symbol, meaning, self or observer may be introduced as primitive. This prevents the theory from hiding the conclusion in its premises. The rule does not guarantee success, but it makes failure visible.

14.3 Recursive reliability-tracking

Many systems model the world. Some systems model themselves. Mirror Theory adds a further requirement: the system must track the reliability of its self-model. This is what separates a basic self-model from recursive observerhood.

14.4 Viability-constrained selection

Mirror Theory does not treat recursion as automatically valuable. Recursive self-modelling is selected only where its predictive or corrective value exceeds its cost. This makes the theory conditional rather than universal.

14.5 A bridge between formal and philosophical work

Mirror Theory is formal enough to state thresholds and failure conditions, but philosophical enough to interpret identity, meaning and objectivity. V01.03's role is to ensure that this bridge does not float in isolation from existing cognitive science.

15 Testable and operational implications

A theory of observerhood should eventually support operational tests. The following implications are framed as research proposals rather than established results.

15.1 Distinguishing feedback from observerhood

Feedback is not observerhood. A thermostat has feedback. A microphone loop has feedback. A recurrent neural network has feedback. Mirror observerhood requires feedback that is organised into self-modelling and reliability-sensitive correction.

Proposal 15.1 (Feedback exclusion test)**.** A system should not be classified as a Mirror observer merely because it contains feedback loops. It must be tested for whether internal representations of its own modelling state influence future prediction, correction or viability preservation.

15.2 Distinguishing self-reference from observerhood

Self-reference is not observerhood. A sentence may refer to itself. A program may print its own source. Such systems lack operational self-maintenance unless the self-referential structure affects prediction, correction or continued organisation.

15.3 Recursive self-model perturbation

If recursive self-model reliability is central to observerhood, then perturbing self-model reliability should produce predictable degradation in observer-like performance.

Proposal 15.2 (Self-model perturbation test)**.** In artificial agents, introduce controlled distortions into self-model variables, reliability estimates or introspective uncertainty estimates. Mirror Theory predicts that agents whose viability depends on recursive self-modelling should show systematic degradation in long-horizon correction, self-maintenance or error recovery when these variables are disrupted.

15.4 Threshold behaviour

Mirror Theory I predicts that recursive observerhood becomes selected when the value of modelling self-model reliability exceeds computational cost. This suggests threshold-like behaviour in artificial systems.

Proposal 15.3 (Recursive threshold experiment)**.** Construct agent classes with increasing access to (i) external state models, (ii) internal state models, and (iii) self-model reliability variables. Measure performance under tasks where future success depends on correcting internal error, uncertainty, fatigue, damage, bias or sensor unreliability. Mirror Theory predicts a performance transition when reliability-modelling becomes action-relevant and cost-effective.

15.5 Artificial observerhood without consciousness claims

Mirror Theory may be especially useful in AI because it offers a way to discuss observer-like organisation without immediately claiming consciousness. An artificial system could be tested for recursive observerhood as a structural property: does it model the world, include itself in the model, and correct the reliability of that self-inclusion over time? That question is more tractable than asking directly whether the system has phenomenal experience.

16 Failure modes

A serious positioning paper must identify the ways the theory could fail. Mirror Theory has several.

16.1 Over-flexibility

If Mirror Theory can describe every adaptive system, it explains nothing. The concept of observerhood must exclude rocks, whirlpools, thermostats, simple feedback controllers and merely self-referential artefacts. The theory must therefore maintain strict architectural thresholds.

16.2 Circularity about symbols

Mirror Theory says that symbols should be derived rather than assumed. But any account of symbolhood risks relying on interpretation, and interpretation may seem to require an observer. The theory must define operational symbolhood through persistent distinctions, counterfactual role, compositional closure and viability relevance rather than human interpretation.

16.3 Phenomenality gap

The formalism may capture recursive observerhood without explaining why experience feels like anything. This is not a rhetorical weakness; it is a genuine unresolved boundary. Mirror Theory should state it openly.

16.4 Subsumption by active inference

It is possible that active inference already contains the useful parts of Mirror Theory once the right generative models and policies are specified. If so, Mirror Theory's contribution may be terminological rather than substantive. To avoid this, Mirror Theory must show that recursive self-model reliability is not merely an optional modelling detail but a distinctive observerhood condition.

16.5 Subsumption by self-model theory

It is also possible that self-model theory already covers the relevant terrain. Mirror Theory must therefore clarify that its contribution is not the claim that selves are models. Its contribution is the derivational and viability-selected account of when self-models become recursively necessary.

16.6 Lack of empirical metrics

Observerhood must eventually be measurable. Without operational metrics for recursive depth, self-model reliability, counterfactual coupling and viability contribution, Mirror Theory remains philosophical architecture.

17 Research programme

The following research programme would turn Mirror Theory from a conceptual framework into a testable programme.

1. **Define operational observerhood metrics.** Develop measures of recursive self-modelling depth, self-model reliability-tracking, counterfactual coupling and viability contribution.

2. **Build artificial agents.** Construct simple agents with and without self-model reliability variables and compare performance on tasks requiring self-correction.
3. **Connect with active inference.** Formalise Mirror observerhood inside active-inference architectures by treating self-model reliability as a hidden state or precision-like variable.
4. **Compare biological cases.** Identify neurocognitive processes in which internal state and self-model reliability affect long-horizon correction, such as metacognition, confidence, body-state estimation and error monitoring.
5. **Develop exclusion tests.** Show why feedback controllers, self-referential programs and simple predictors fail the observerhood condition.
6. **Clarify relation to phenomenology.** Determine whether recursive observerhood is merely a structural prerequisite for consciousness or a partial explanation of it.
7. **Specify minimal substrates.** Investigate what kinds of physical, biological or artificial systems can instantiate the architecture.

18 Conclusion

Mirror Theory should not be sold as a replacement for the existing literature. Its value lies in a narrower and more disciplined proposal: observerhood may be a derivable computational regime rather than a primitive subject or unexplained point of view.

This paper has argued that Mirror Theory sits between several traditions. From predictive processing and active inference, it inherits modelling, prediction and correction. From autopoiesis and enactivism, it inherits self-maintenance, embodiment and world-making through engagement. From global workspace theory, it acknowledges the importance of availability and integration. From IIT, it learns the value of formal seriousness about consciousness while rejecting consciousness-first methodology. From self-model theory, it inherits the idea that the self is modelled, while adding viability-constrained recursive reliability-tracking. From Dennett and Hofstadter, it inherits anti-Cartesian and recursive insights while refusing to equate external attribution or bare self-reference with observerhood.

The central thesis can now be stated in one paragraph.

> *A system becomes a Mirror observer not merely by processing information, predicting inputs, maintaining feedback, being alive, globally broadcasting content, integrating information, referring to itself or being interpreted as intentional. It becomes a Mirror observer when it maintains a compressed model of the world, includes itself within that model, and tracks the reliability of that self-inclusion in a way that matters for prediction, correction and continued organisation.*

If this is correct, Mirror Theory's contribution is not to settle consciousness, but to shift the order of inquiry. Before asking what consciousness is, we ask what kind of system can become an observer. Before asking whether the self is real, we ask what organisation maintains a self-model. Before asking whether reality is mind-dependent, we ask how constraints become a world for a system. That is the narrow but potentially important place of Mirror Theory in context.

A Comparison matrix

Table 2: Comparison of Mirror Theory with neighbouring frameworks.

Framework	**Observer derived?**	**Self-model required?**	**Reliability tracking required?**	**Phenomenality solved?**	**Mirror Theory distinction**
Predictive processing	No, generally assumed as system or agent	Not necessarily	Sometimes via precision or uncertainty, but not observer-defining	No	Adds observerhood criterion to predictive architectures.
Active inference	Partly, as agent-environment system	Optional depending on model	Possible via precision or meta-models	No	Identifies recursive self-model reliability as observerhood threshold.
Autopoiesis	Living unity derived	Not necessarily	No	No	Life/autonomy not sufficient for observerhood.
Enactivism	Partly via embodied sense-making	Not always explicit	Not central	No	Adds formal recursive self-modelling condition.
Global workspace	No, focuses on access	Not required by base theory	No	Addresses access more than observer derivation	Mirror asks what architecture makes global access observer-relevant.
IIT	Conscious substrate identified from axioms	Not required	No	Intended as theory of consciousness	Mirror begins with observerhood, not phenomenology.
Self-model theory	Self-model analysed	Yes	Not central as viability-selected recursion	Addresses phenomenal self-model	Mirror derives why self-model reliability matters.
Intentional stance	Observer attributed by interpreter	No	No	No	Mirror seeks internal constructive conditions.
Strange loops	Self-reference central	Often symbolic self-reference	Not viability-based	Partial philosophical account	Mirror distinguishes bare recursion from operational observerhood.
Mirror Theory	Yes, under conditions	Yes	Yes	No, explicitly open	Recursive observerhood as derived regime.

B Glossary of key terms

Observerhood.
The structural regime in which a bounded system maintains a world-model that includes the system itself and the reliability of its own modelling activity.

Viability. Organisational preservation; the continued maintenance of the system as the relevant kind of system. Not moral value.

Compression.
The selective preservation of differences relevant to prediction, action or viability under bounded capacity.

Self-model. A model of internal or system-relative variables that affect future prediction, action or persistence.

Recursive self-model.
A model that includes not only internal state but the reliability, error-profile or adequacy of the self-model itself.

Worldhood. Reality as it appears to a system through interpreted, compressed and action-relevant modelling.

Constraint. Structure that resists arbitrary reinterpretation by affecting prediction, action, coordination or viability.

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